ROBOTIC PROCESS AUTOMATION IN INNOVATION MANAGEMENT: A COMPARATIVE ANALYSIS ACROSS PROCESS FRAMEWORKS

Lappeenranta–Lahti University of Technology LUT

Master's Thesis in Global Management of Innovation and Technology

2025

Timi Marviala

Examiners: Professor Ville Ojanen

Associate Professor Kalle Elfvengren

ABSTRACT

Lappeenranta–Lahti University of Technology LUT

School of Engineering Science

Industrial Engineering and Management

Timi Marviala

ROBOTIC PROCESS AUTOMATION IN INNOVATION MANAGEMENT: A COMPARATIVE ANALYSIS ACROSS PROCESS FRAMEWORKS

Master's thesis

2025

63 pages, 4 figures, 5 tables

Examiners: Professor Ville Ojanen and Associate Professor Kalle Elfvengren

Keywords: Robotic Process Automation (RPA), Automation, Innovation Management, Innovation Process, Open Innovation, Agile Innovation, Lean Startup, Design Thinking, TRIZ, Stage-Gate Model

This thesis examines where Robotic Process Automation (RPA) can deliver measurable value within different innovation process frameworks. While RPA is widely used in business operations, its role in innovation management has been unclear. The study addresses this gap by analyzing six frameworks to see where RPA could best be implemented and what benefits it could bring. The innovation frameworks analyzed are Open Innovation, Design Thinking, Lean Startup, Agile Innovation, TRIZ, and the Stage-Gate model.

The research is based on a qualitative study of academic literature, professional reports, and vendor cases. Each framework is broken down into stages and evaluated. The results show that RPA provides the most value in stages that are structured, rule-based, and produce standardized outputs. The strongest automation opportunities lie in reducing cycle time, improving accuracy, and strengthening auditability. The thesis discusses the results by linking RPA's role to a work profile characterized by repeatability and structure and a practical roadmap is proposed for implementation. The conclusions underline that automation can reinforce innovation processes by taking over documentation-heavy tasks, but further research should explore its interaction with creativity and the potential of more advanced automation technologies.

TIIVISTELMÄ

Lappeenrannan–Lahden teknillinen yliopisto LUT

LUT Teknis-luonnontieteellinen

Tuotantotalous

Timi Marviala

OHJELMISTOROBOTIIKKA OSANA INNOVAATIOJOHTAMISTA: VERTAILEVA ANALYYSI ERI PROSESSIMALLEISTA

Tuotantotalouden diplomityö

2025

63 sivua, 4 kuvaa, 5 taulukkoa

Tarkastajat: Professori Ville Ojanen ja Dosentti Kalle Elfvengren

Avainsanat: Ohjelmistorobotiikka (RPA), Automaatio, Innovaatiojohtaminen, Innovaatioprosessi, Avoin Innovaatio, Ketterä Innovaatio, Lean Startup, Muotoiluajattelu, TRIZ, Stage-Gate -Malli

Tämä diplomityö tarkastelee, missä kohdin erilaisia innovaatioprosessien työkaluja ohjelmistorobotiikka (RPA) voi tuottaa mitattavaa arvoa. Vaikka RPA on laajasti käytössä erilaisissa liiketoiminnan operatiivisissa prosesseissa, sen rooli innovaation tukena on epäselvä. Tutkimus paikkaa tätä aukkoa analysoimalla kuutta innovaatiomallia tunnistaakseen mihin vaiheisiin RPA soveltuu parhaiten ja mitä hyötyjä siitä voisi olla. Analysoidut innovaatiomallit ovat avoin innovaatio, muotoiluajatteluprosessi (Design Thinking), Lean Startup, Agile Innovaatio, TRIZ sekä Stage-Gate-malli.

Tutkimus perustuu kvalitatiiviseen tarkasteluun akateemisesta kirjallisuudesta, asiantuntijaraporteista ja toimittajien tapaustutkimuksista. Kukin innovaatiomalli on jaettu vaiheisiin ja arvioitu erikseen. Tutkimuksen tulokset osoittavat, että RPA tuottaa eniten arvoa vaiheissa, jotka ovat selkeästi määriteltyjä, sääntöihin perustuvia ja tuottavat standardoituja tuotoksia. Suurimmat hyödyt liittyvät prosessien nopeutumiseen, tarkkuuden paranemiseen sekä auditointimahdollisuuksien vahvistumiseen. Työn lopussa esitetään myös käytännönläheinen etenemissuunnitelma sen käyttöönotolle innovaatioprosesseissa. Johtopäätöksissä korostetaan, että automaatio voi vahvistaa innovaatioprosesseja hoitamalla dokumentaatiovetoisia tehtäviä, mutta jatkotutkimuksessa tulisi kuitenkin tarkastella sen yhteyttä myös luoviin vaiheisiin sekä siirtymistä kehittyneempiin automaatioteknologioihin.

ACKNOWLEDGEMENTS

I would like to thank everyone involved in this work, starting with my research supervisor Ville Ojanen, who guided me through the writing process. Additionally, the whole academic staff of LUT University deserves my gratitude, especially all the lecturers from the Global Management of Innovation and Technology Master's program.

I am also grateful to the people I work with outside of LUT, especially the RPA specialist whose invaluable insights and feedback helped me connect this thesis to real-world practice. Lastly, I would like to thank my mother for her constant encouragement and support throughout my studies.

ABBREVIATIONS

AI	Artificial Intelligence
CRM	Customer Relationship Management
IP	Intellectual Property
MVP	Minimum Viable Product
RPA	Robotic Process Automation
R&D	Research and Development
TRIZ	Theory of Inventive Problem Solving
UX	User Experience

Table of Contents

Abstract

Tiivistelmä

Acknowledgements

Abbreviations

Declarations

1 Introduction 9
1.1 Research Objectives and Questions 9
1.2 Scope 10
1.3 Methodology and Data Collection 11
1.4 Research Structure 12
2 Innovation Processes and Frameworks 13
2.1 Open Innovation 13
2.2 Flexible Iterative Innovation 15
2.2.1 Design Thinking 15
2.2.2 Lean Startup 17
2.2.3 Agile Innovation 19
2.3 Structured Innovation 19
2.3.1 TRIZ 19
2.3.2 Stage-Gate Model 20
3 Robotic Process Automation 23
3.1 Definition of RPA 23
3.2 Current Use Cases of RPA in Business 24
3.3 Benefits and Limitations 26
3.3.1 Benefits 26
3.3.2 Limitations 28
3.4 Role of RPA in Innovation 29
4 Analysis 31
4.1 Evaluation Criteria 31
4.2 Open Innovation 32
4.3 Design Thinking 35

4.4 Lean & Agile Innovation 37
4.5 TRIZ 39
4.6 Stage-Gate 42
4.7 Cross-Framework Comparison 45
5 Discussion 47
5.1 Ease of RPA Implementation in Innovation 47
5.2 Measurable Benefits of RPA in Innovation 48
5.3 Value of RPA in Innovation 50
5.4 Theoretical Implications of the Results 52
5.5 Validity, Limitations and Transferability of the Results 53
5.6 Implementation Roadmap 54
6 Conclusions and Future Research 57
References 59

DECLARATIONS

Turnitin

The originality of this thesis has been reviewed with the Turnitin similarity checking service.

AI usage

The author of the thesis, Timi Marviala, used the following AI-tools during the preparation of the thesis:

1. ChatGPT
    a. Purpose of use: For brainstorming, paraphrasing, spelling correction, relevance checks.
    b. Explanation of the use of the tool: This tool helped with ideation and planning of what should be included in this study. The tool was also sometimes used to help improve sentence structure and spelling if needed. Additionally, the tool was used for quick relevance checks and citation chaining before delving into full-text access in certain sources.

Responsibility

The author, Timi Marviala, takes full responsibility for the content of this thesis and has reviewed and edited the content generated by the possible use of AI tools.

1 Introduction

Innovation is a key driver of competitiveness, yet in many organizations it is still treated mainly as an outcome instead of a managed process. When innovation is viewed this way, the supporting activities, roles and tools inside the process are not systematically developed. This creates blind spots for operational improvements that could strengthen innovation performance over time.

At the same time, Robotic Process Automation (RPA) has become a common way to improve efficiency in business operations by automating repetitive, rule-based tasks. These effects can release people from routine work to focus on analysis and creative tasks. These conditions should, in principle, support innovation, however most accounts still describe RPA's connection to innovation indirectly through productivity gains rather than as a direct element inside innovation management.

The motivation for this thesis is the observed gap between these two areas. While RPA is widely used in operational processes, there is little examination of where exactly inside established innovation frameworks RPA could be integrated and what value that could create for each stage. Existing texts acknowledge the indirect benefits, but there is limited guidance for managers who need to decide which phases of their innovation process are both feasible and worthwhile to automate with RPA.

1.1 Research Objectives and Questions

While RPA already has a huge role in supporting processes by automating routine tasks across industries, there is a gap when it comes to innovation management. Since innovation is still often seen as an outcome, rather than a process (Garud, Tuertscher & Van de Ven 2013), it is likely that companies don't yet fully understand the potential robotic process automation could offer. The existing literature suggests that RPA can drive innovation indirectly by freeing up employees for more creative work (Lacity, Willcocks & Craig 2015), yet there is little to no examination on where RPA can be integrated within specific innovation frameworks.

The aim of this thesis is to address that gap by analyzing which parts of different innovation processes are most suitable for RPA implementation and how automation can benefit said processes. The objective is to identify key areas of innovation that would benefit the most from automation and compare those to areas that RPA would be easiest to implement. The end goal is to answer the sub research questions:

"How easily can RPA be implemented to each stage of the established innovation process frameworks?"

"What measurable benefits does RPA bring to each stage of the of the established innovation process frameworks?"

The combined analysis of these sub questions form the answer to the main research question:

"Where, within established innovation process frameworks, can Robotic Process Automation (RPA) deliver measurable value?

1.2 Scope

Although numerous frameworks exist to facilitate innovation management, this thesis focuses specifically on six widely recognized and extensively adopted models: Open Innovation, Design Thinking, Lean Startup, Agile Innovation, Stage-Gate and TRIZ. These frameworks were selected based on conceptual diversity, structural clarity and maturity as well as their wide usage across industries. The selected frameworks vary from more flexible iterative methods to structured and systematic approaches. This variety ensures a comprehensive evaluation of the potential of RPA.

Additionally, the selected models are sufficiently mature and well documented in academic literature, enabling reliable definitions and a meaningful analysis. Innovation methods that serve a more niche purpose or lack the structured repeatable processes that are essential to clearly identify automation opportunities are excluded from the scope.

In this thesis, for all intents and purposes, Robotic Process Automation (RPA) is defined as a technology that uses software robots to automate rule-based tasks typically performed by humans (van der Aalst et al. 2018). The thesis does not cover each RPA tool separately, but

the technology as a whole, since the use cases of different RPA tools are more or less the same. AI-based automation is also excluded from the scope as the focus is not to replace creative ideation but support other parts of innovation processes.

While RPA has been extensively studied in areas such as finance, HR, supply chain, and customer service, its role inside established innovation frameworks has received little academic attention. Existing research usually describes RPA's impact indirectly, but does not map where in the innovation process RPA could be most directly valuable. This thesis addresses that gap by conducting a comparative analysis across the selected six frameworks.

1.3 Methodology and Data Collection

This thesis applies a qualitative, exploratory approach, which is often utilized when a study aims to develop an in-depth understanding of a poorly defined or emerging phenomenon by identifying underlying patterns, relationships, and meanings rather than testing predefined hypotheses (Creswell & Creswell 2017). The topic of identifying where in the established innovation RPA can be integrated, has limited prior quantification, so a qualitative design is appropriate to generate structured, practical insights. The objective is to provide actionable guidance and opportunities for RPA inside well-known innovation frameworks.

As a qualitative study, data collection for the thesis was desk research conducted using trusted discovery tools. The search began in LUT Primo and Google Scholar to retrieve books and peer-reviewed articles across licensed collections. Search queries combined framework names (e.g. "Stage-Gate", "Lean Startup", "TRIZ") and stage terms with RPA terminology (e.g. "automation", "process automation", "RPA in innovation") and were refined iteratively based on initial hit quality. Priority was given to more recent publications since RPA as a technology is fairly new. Inclusion criteria required that sources addressed RPA, business process automation or innovation management frameworks. Professional reports were only included if they came from well-established consulting firms, recognized professional organizations, or widely cited industry analysts. Exclusion criteria removed sources that focused on artificial intelligence as the scope of this thesis is limited to UI-level RPA. For the more practice-oriented material, professional organization websites and credible industry reports were consulted. Any relevant vendor white papers were cross-checked

against neutral descriptions. The work uses published materials only and no personal data was collected.

In later analysis each innovation framework mentioned earlier is decomposed into its canonical stages defined in chapter 2 and for each stage, two criteria are assessed: Ease of implementation of RPA, and perceived benefits from such automation. Organizational readiness factors are deliberately excluded to keep the evaluation broadly applicable across firms and industries. While the frameworks are assessed in a uniform way and justifications are documented, the results still reflect expert judgement by a single researcher and thus involve subjectivity. These results should be interpreted as structured qualitative assessments rather than measurements.

1.4 Research Structure

The study follows a straightforward progression from theory to application. It begins in chapter 2 by establishing a foundation in existing innovation management literature, where six innovation frameworks are reviewed and compared to illustrate how structured processes guide innovation work. Building on this, chapter 3 then introduces Robotic Process Automation as a technological concept, outlining its operating principles and summarizing prior research on its business applications.

After defining both perspectives, chapter 4 analyzes the integration of RPA within these innovation frameworks. Each framework is examined to identify stages and activities that align with RPA's technical strengths, allowing a structured comparison of automation opportunities across different process models. The findings are then reflected in chapter 5 against earlier studies to highlight theoretical implications and practical relevance, followed by a synthesis that proposes a concrete roadmap for implementing RPA in innovation management. The study concludes by summarizing the key insights, limitations and opportunities for further research.

2 Innovation Processes and Frameworks

Innovation is a key factor in sustaining competitiveness and driving growth in any organization. To effectively utilize creativity while managing risk and execution, businesses increasingly rely on processes. Over time, a variety of innovation frameworks and methodologies have been developed to support the ideation, development, and implementation of new products and services. These frameworks offer established workflows, decision-making tools, and best practices that help organizations maximize innovation outcomes while minimizing uncertainty and inefficiencies.

This chapter provides an overview of six widely recognized innovation frameworks, examining their core principles, strengths and limitations. Each process is divided into phases to lay the groundwork for identifying which parts of innovation may be best suited for integration with RPA in later chapters. The frameworks are introduced in order of increasing structural formality, beginning with Open Innovation, followed by a group of user-centric, iterative models and ending with more structured approaches. It is important to mention that these frameworks are not mutually exclusive and in practice, many organizations apply multiple methodologies together depending on the context and objectives.

2.1 Open Innovation

Open Innovation is a framework that in addition to internal R&D, also promotes the use of external knowledge sources to drive innovation. In more traditional innovation models, companies rely solely on internal capabilities to generate new ideas. Open Innovation, however, suggests that collaboration with customers, suppliers, research institutions, startups and even competitors can help accelerate innovation and expand market opportunities as well as reduce risks. (Chesbrough 2003)

The core principles of Open Innovation include both inbound and outbound innovation as well as collaborative networks. This means that companies can leverage external sources of ideas and intellectual property to help with their own R&D but also allow their internal ideas to be commercialized outbound, especially when they don’t fit the company’s strategy

anymore. This allows for cross organizational partnerships, ecosystems and even platforms to share knowledge and co-create (Chesbrough 2003). As a less structured model, open innovation doesn't have a linear workflow with set phases. However, in the context of this study the framework is divided into distinct phases for easier systematic analysis. These phases are idea sourcing, idea integration, idea commercialization and collaboration management.

Open Innovation can provide multiple significant advantages that have made it a popular model across industries. By opening the innovation process, companies can tap into knowledge beyond their own capabilities, accelerating problem solving, reducing development time and even costs. This enhanced innovation capacity often also leads to more disruptive and user-driven innovations. A good example of an Open Innovation success story is the Connect and Develop platform created by Procter & Gamble. The platform invited innovators and inventors to contribute ideas, which later resulted in over 50% of Procter & Gamble's new products to be credited to external sources (Huston & Sakkab 2006).

However, there are also plenty of challenges that rise with Open Innovation. The first and most obvious one being the risks related to IP. Sharing information with external stakeholders can raise concerns about IP theft, confidentiality and the misalignment of incentives between the parties involved. Relying on external partners also reduces the company's direct control over the innovation pipeline, which can lead to the brand and strategy drifting.

There could also be problems with idea integration between organizations. Since external ideas often come with different standards, it can be harder to fit them into existing R&D (Chesbrough 2003). Managing partnerships and external collaborations can require more resources and governing mechanisms (Huston & Sakkab 2006). Because of this, Open Innovation tends to work best when organizations already have clear strategic goals and well-defined IP management.

2.2 Flexible Iterative Innovation

This section discusses flexible iterative innovation, which in the context of this thesis includes innovation approaches that cycle between learning and doing. The chosen iterative processes are Design Thinking, Lean Startup and Agile Innovation.

2.2.1 Design Thinking

Design Thinking is a human-centered problem-solving approach that's deeply rooted in the practices of designers but also utilized in product design, business strategy, healthcare, education and digital transformation, making it one of the most influential innovation frameworks of all time (Brown 2008). Design Thinking is built on the core principles of empathy, ideation, prototyping and implementation with the idea of integrating peoples needs, technological feasibility and business viability (Liedtka 2015).

These principles emphasize a non-linear and iterative process meaning that teams often cycle between different phases as they learn more about the problem and potential solutions. While Design Thinking is typically understood as a flexible and iterative methodology rather than a rigid step-by-step process, it is often visualized through models that break it into distinct phases to support application and analysis. A good example of this is the widely recognized framework for Design Thinking called the double diamond model developed by the UK Design Council. This model has four stages: discover, define, develop and deliver (Design Council 2004) and since it provides a clear, widely accepted and practical framework, it will be used to represent Design Thinking as a whole in the context of this thesis. This model is represented in figure 1 below.

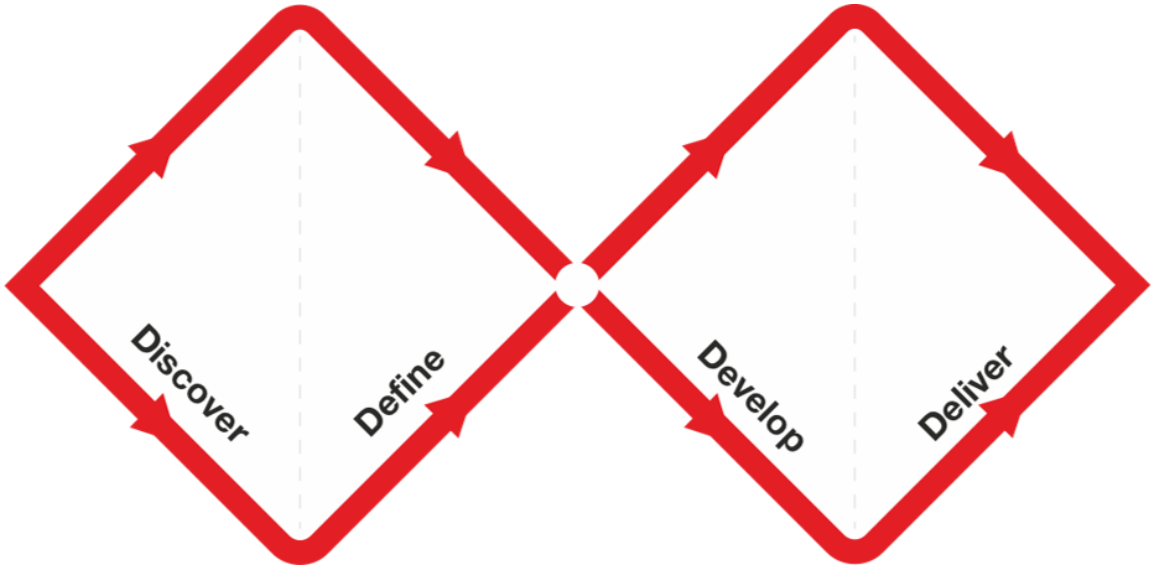


Figure 1: The Double Diamond (Design Council 2004)

In the Discover phase, the goal is to understand the user as well as the problem by gathering insights through research and observation. This is followed by the Define phase, where those insights are synthesized into a clear and actionable problem statement. The Develop phase focuses on generating ideas and building prototypes, encouraging experimentation and iteration to explore possible solutions. In the Deliver phase, the most promising solutions are improved and implemented based on user feedback, ensuring the outcome is usable and valuable. (Design Council 2004)

Design Thinking is widely regarded as an effective innovation tool due to its user-centric focus, which ensures that solutions align with real user needs and expectations. This can be seen in multiple real-life examples, with many technology companies such as Apple (Thomke & Feinberg 2009) using Design Thinking to enhance UX and develop human-centered products. The framework also encourages creativity and allows for quick testing and iteration, which in turn can reduce development risks. This can be especially important with complex technological and software products. By taking a human-centric approach, Design Thinking can be seen as tool for simplifying complex problems. Because of this industry giants such as IBM have incorporated Design Thinking into its corporate strategy improving collaboration across teams and accelerating product innovation cycles (Kolko 2015).

Despite its advantages, Design Thinking can be seen to have several challenges as well (Kolko 2015). Since it's a flexible model it's harder to follow strict decision points. The iterative nature and flexible guidelines also make it fairly challenging to quantify short-term business impact. This can also lead to the model being harder to effectively scale across larger organizations with multiple departments for example. These limitations would suggest that Design Thinking would be best implemented with a combination of other tools or frameworks to balance creativity with quantifiable results.

2.2.2 Lean Startup

The Lean Startup methodology is a modern innovation framework focused on developing products and businesses through rapid experimentation, validated learning and iterative development (Ries 2011). Heavily inspired by lean manufacturing principles, the model has gained popularity among startups and innovation-driven teams operating in fast technological environments (Blank 2013). The core principle of Lean Startup is to minimize waste by avoiding unnecessary development (Ries 2011).

Using Minimum Viable Product, or MVP for short, companies can test a version of the product with just the necessary features. Then, using real customer feedback from the MVP, it can be defined whether the idea solves a meaningful problem. Based on a continuous feedback loop, development teams can then decide if they should continue in the same direction or try a new approach. Lean Startup is not a process set in stone, but rather a mindset and framework designed to encourage agility especially in the early stages of product development, however it does have a distinct set of phases represented in Figure 2 below. (Ries 2011; Blank 2013)

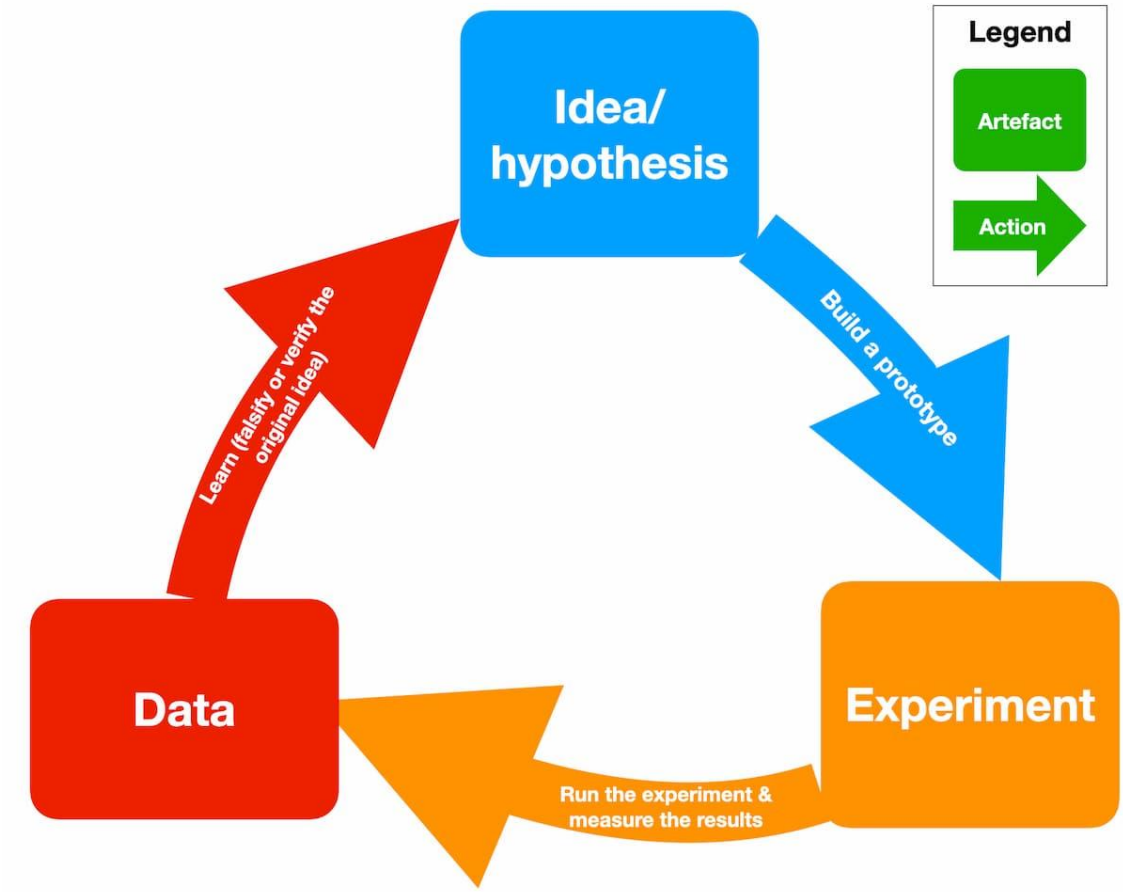


Figure 2: Lean Startup (Antonsk 2021

The Lean Startup model offers several benefits, especially in situations where the innovated product is disruptive, and the market is unknown (Blank 2013). Real-time customer feedback guiding the decisions will lead to a user-centric end-result while allowing the company to avoid long development cycles with uncertain outcomes. Small iterative experiments help teams fail early and cheaply, making the model flexible and low risk. These benefits have got even larger corporations like GE to utilize it as a part of their innovation initiatives (Blank 2013).

However, Lean Startup is not without its challenges. It can be difficult to utilize it in highly regulated and hardware dependent industries since creating MVPs may be expensive or even impossible. Also, with innovators focused on speed and short-term feedback loops, the bigger picture can be missed. Customer feedback can even be wrong, since early adopters might not represent the broader market. (Ries 2011)

2.2.3 Agile Innovation

Agile Innovation is an approach that combines the principles of agile project management with the goals of innovation. Originally rooted in software development (Beck et al. 2001), agile innovation has since been adopted more broadly across various functions in business and R&D (Rigby, Sutherland & Takeuchi 2016). Similarly to Lean Startup, the key idea of Agile Innovation is to embrace change and respond quickly to user feedback. Cross-functional teams working in short development cycles allow for ideas to be rapidly tested and adjusted.

In practice, Agile Innovation follows a structure that is very close to Lean Startup. Both of them emphasize iterative loops, real-time customer feedback and continuous adaptation. Because of this similarity, many of the advantages and limitations associated with Lean Startup also apply to Agile Innovation. These include the ability to reduce waste, accelerate learning, and quickly respond to uncertainty but also challenges in long-term planning, strategic alignment, and applicability in highly regulated industries. Due to this overlap in structure these two models will be analyzed together later in this study.

2.3 Structured Innovation

This section moves to the structured side of innovation with Stage-Gate and TRIZ. Stage-Gate organizes development into phases separated by explicit decision points and document sets while TRIZ offers a systematic toolkit for moving from problems to candidate solutions by resolving contradictions. Both aim to reduce uncertainty, but they do it in different ways.

2.3.1 TRIZ

Theory of Inventive Problem Solving, or TRIZ for short, is a structured innovation methodology developed in the Soviet Union starting as far back as the 1940s. Unlike more exploratory frameworks, the main idea of TRIZ is that inventive solutions follow repeatable patterns that can be applied across different products and fields (Altshuller 1984). In short, TRIZ transforms innovation from unpredictable creativity into a repeatable and systematic process.

TRIZ consists of several tools designed to support structured problem solving. These include the 40 Inventive Principles, the Contradiction Matrix, Trends of Technological Evolution, Substance-Field Analysis, and the Algorithm for Inventive Problem Solving (Altshuller 1999). Although TRIZ is commonly viewed as more of a toolbox, the application of said tools can be divided into phases to support structured analysis. Firstly, the problem needs to be defined with specific technical or functional terms. After this, conflicting requirements or contradictions such as improving strength without increasing weight are identified. Then the problem is translated into a higher-level description that is free from domain-specific constraints. Solutions can then be found for this higher-level description by analyzing patents in other domains for example. The focus of this stage is to find as many solutions as possible and then later evaluate and refine these general solutions based on their feasibility. (Gadd 2011).

TRIZ has several key advantages. It is effective in technical problem-solving environments, such as engineering and manufacturing, where innovation often involves working within physical constraints. The structured nature of TRIZ allows it to reduce psychological inertia, since users are encouraged to think beyond familiar solutions by applying abstract principles (Cavallucci & Rousselot 2011). It also excels at addressing contradictions systematically, which makes it a valuable method for generating breakthrough solutions rather than incremental improvements.

However, TRIZ is not without limitations. One common criticism is the learning curve. Understanding and applying all the TRIZ tools requires training and practice (Moehrle 2005). It can also be perceived as rigid or theoretical, especially in domains where the abstraction process may feel disconnected from user needs. In addition, while TRIZ doesn't offer much guidance on how to implement, test or scale innovations, meaning it is often used best in combination with more iterative or user-centered approaches such as Design Thinking or Lean Startup (Gadd 2011).

2.3.2 Stage-Gate Model

The Stage-Gate model is one of the most widely used structured innovation frameworks in corporate R&D environments. The model was designed to improve the management of product development projects by introducing a step-by-step structure with clear decision points.

Its main purpose is to reduce uncertainty, improve project visibility, and ensure that development resources are only committed to projects with validated potential (Cooper 1990; Cooper 2008).

The Stage-Gate process is divided into a series of stages and gates which act as decision points to assess progress and determine whether a project should continue or be redirected or even terminated. While there may be some variety in the numbering and naming of the stages depending on organization, a typical Stage-Gate process includes five stages presented in figure 3 below: scoping, business case development, development, testing/validation, and launch. Each stage involves a set of defined tasks and deliverables, such as market research, technical analysis, prototyping, or customer validation. The gates between stages are structured reviews, where a team evaluates the project based on criteria like strategic fit, market attractiveness, technical feasibility or financial return. The gatekeepers then decide before allowing the project to move forward (Cooper 2011).

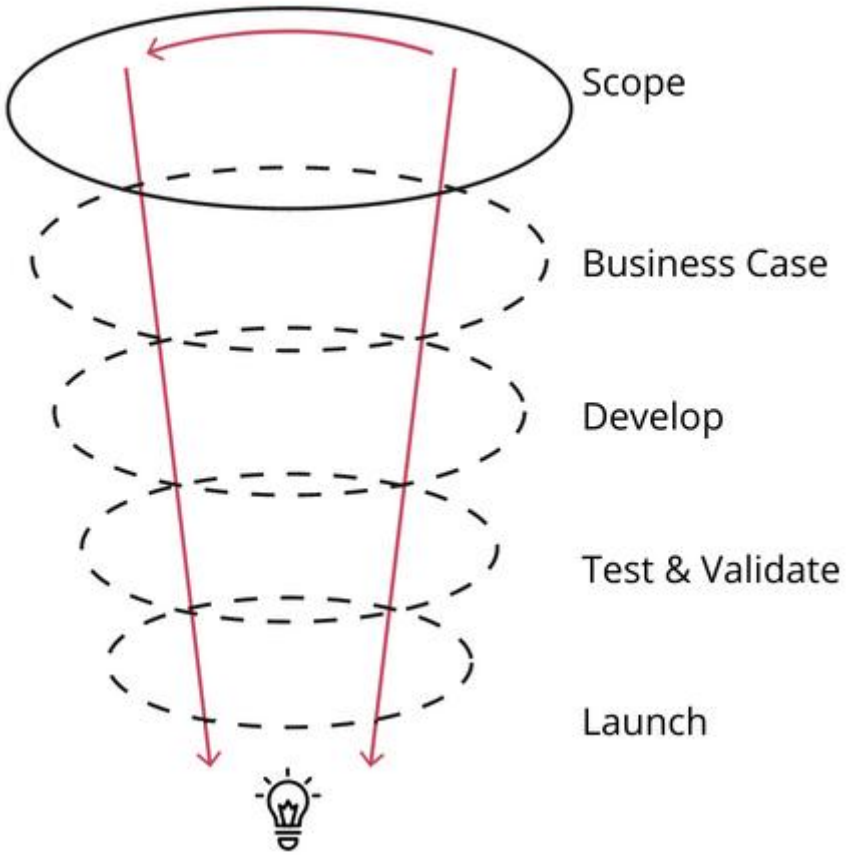


Figure 3: Stage-Gate model (Slade74 2022, Cooper 1990)

The key advantage of the Stage-Gate model is its ability to bring structure and risk control to innovation projects. This is especially helpful in large organizations with complex product portfolios. The model ensures that only high potential ideas move forward and resources aren't wasted on the wrong things. The staged approach also supports cross-functional coordination, clearer roles and responsibilities, and early identification of critical issues.

On the other hand, the Stage-Gate model has also received criticism for being overly rigid and slow adapting. Strict gate reviews can discourage innovators and delay the time it takes to get a product on the market. In response to these concerns, many companies have adapted their Stage-Gate models to be more flexible, incorporating elements from agile or lean approaches with the aim to preserve benefits of Stage-Gate while increasing adaptability and speed (Cooper & Sommer 2016a).

3 Robotic Process Automation

This chapter defines what RPA means in this thesis and sets the scope. The focus being on UI-Level RPA or bots that follow clear rules to operate existing tools via their interfaces. With these boundaries set, the chapter provides the technical baseline used in later analysis to judge where RPA can be implemented reliably and where it can add measurable value.

3.1 Definition of RPA

Robotic Process Automation (RPA) can broadly be defined as a form of business process automation that uses software "robots" to perform tasks by emulating human interaction with digital systems (van der Aalst et al. 2018). RPA involves programs that replicate the actions of a human user within computer applications, executing operations such as clicking buttons, entering data, or copying information between systems. RPA is not a form of artificial intelligence, since the bots follow predefined rules and workflows and do not learn or make autonomous decisions (Redwood 2024). RPA bots can operate either in an attended mode or in an unattended mode and this flexibility allows RPA to be deployed in different scenarios, from desktop assistants that help individual employees to back-end processes that are executed automatically.

A core characteristic that differentiates RPA from traditional automation approaches is how the automation is implemented. Conventional automation typically requires a software developer to write code to integrate with each application. In such cases, automating a process usually involves modifying the underlying software or database, or developing custom interfaces between systems. RPA, by contrast, takes a different approach: it works through a system's existing user interface rather than altering the system's code (IBM 2023). An RPA tool can observe a user performing a task step by step in an application and then generate a script that allows the software robot to reproduce those same steps. For example, to automate an invoice processing task, an RPA bot might be programmed to open an email, copy an invoice number, paste it into an accounting system's form, and click through screens to update a record. In other words, RPA operates at the presentation layer of software and does not require deep integration with the underlying databases of each application. This means

organizations can implement RPA on top of 3rd party systems, which significantly lowers the barrier to automation in environments where traditional integration is costly or impractical.

Another fundamental aspect of RPA is the accessibility of its development. RPA platforms are typically designed to be no-code or low-code, meaning that building a bot does not require advanced programming skills. Users define the logic of a process either by using visual workflow designers or by recording their actions as they perform a task, which the RPA software then converts into an executable script (UiPath 2025). Modern RPA tools come with intuitive, low-code capabilities to configure automation scripts, along with features to integrate with enterprise applications and to orchestrate and monitor the bots' activities (UiPath 2023). The result is a faster development cycle for automation solutions and the ability for organizations to iterate quickly on process improvements. In summary, the combination of working through the UI and offering a low-code development environment makes RPA a flexible and user-friendly automation technology compared to earlier generations of enterprise automation tools.

RPA is best suited for tasks and processes that are rule-based, repetitive, and involve structured data. Typical RPA candidates are those back-office functions that follow a well-defined procedure and do not require human judgment or complex decision-making at each step. By programming a software robot to handle such work, organizations can execute these processes with high speed and accuracy, and with consistent adherence to the prescribed rules. In these scenarios, an RPA bot can reliably follow the predetermined workflow from start to finish. (Baker Tilly 2023)

3.2 Current Use Cases of RPA in Business

Robotic Process Automation is being utilized across a wide range of business functions wherever work is high-volume, repetitive, and rules-based. In practice, RPA is most often applied in finance and accounting, human resources, customer service/CRM, as well as supply chain & logistics functions.

One of the leading adopters of RPA is the finance and accounting field, since it involves many routine tasks that must be executed with accuracy, such as invoice processing, account

management, expense reporting, and other financial reporting. Software bots can extract invoice data, post journal entries, reconcile discrepancies, and compile reports much faster and with fewer errors than manual processing. For example, Jaguar Land Rover (JLR) implemented RPA to automate its accounts payable and invoice reconciliation processes, which increased processing accuracy and reduced compliance risks by eliminating manual effort (Blue Prism, 2025a). This RPA deployment at JLR freed up significant human capacity leading to savings that well exemplify how RPA is streamlining finance operations by taking over labor-intensive tasks.

Human Resources departments are also leveraging RPA to handle repetitive administrative work so that HR staff can focus on more people-centric activities. Typical HR processes suited for RPA include employee onboarding and offboarding, payroll processing, time and attendance reporting as well as employee data management. RPA bots in HR can automatically collect and enter employee information across systems, schedule interview appointments, generate contract documents, or update payroll changes according to predefined rules. A notable example is Santander Bank's HR function, which used RPA to automate the new-hire onboarding process. The RPA bots automatically notify all relevant departments when a new employee is added, gather required documentation, set up user accounts, and perform necessary compliance checks. As a result, Santander reduced its onboarding time from six weeks to just two days while greatly improving the new employee experience (Blue Prism, 2025b). This illustrates how RPA delivers speed and consistency in HR services by handling the high-volume transactional tasks that were once done manually.

RPA is also used to support customer service and customer relationship management by automating routine service requests and data updates behind the scenes. Frequent use cases in this area include processing customer orders, updating customer account information in CRM systems, handling basic service inquiries, migrating data between customer service platforms, and generating routine communications. By doing these repetitive tasks with bots, companies ensure that customer requests are handled faster and with fewer errors, allowing human agents to concentrate on complex customer needs. For instance, in the telecommunications sector, Telefónica O2 (UK) deployed an RPA program at scale for its customer operations. The company introduced over 160 software robots to process around 500,000 customer transactions each month (Lacity, Willcocks & Craig 2015). This large-scale automation clearly helped Telefónica O2 deliver services more rapidly to customers. Another

example of using RPA to speed up claims handling comes from the insurance company Zurich UK, who automated more than 120 insurance processes with RPA dramatically reducing the time customers had to wait for claims payouts (Blue Prism, 2025c).

Beyond front-office functions, RPA is also widely applied in operational processes such as supply chain management, procurement, and logistics. Many of these processes involve large volumes of data and coordination across systems, making them ideal for automation. For example, RPA bots can automatically extract order details from purchase orders, enter shipments into tracking systems, or reconcile shipping documents. A notable case is DHL Global Forwarding, which created a centralized RPA "Virtual Delivery Center" in its shared services operations to streamline internal logistics and finance workflows. Within the first year, DHL deployed over 80 RPA robots, which now handle work equivalent to 300 full-time employees' workload in back-office processes. This massive automation initiative removed process bottlenecks and allowed DHL's human workforce to focus on more value-adding activities. (UiPath 2025)

Outside of the core functions above, organizations in various sectors can also find more niche applications for RPA. In the public sector, government agencies can use RPA to improve administrative efficiency. In healthcare, hospitals can introduce RPA for patient scheduling, billing, and transcribing data from one system to another. Even IT departments can utilize RPA for routine support tasks like password resets for example. All this highlights RPA's flexibility, since current use cases of RPA in business range from finance and HR to customer service, logistics, and beyond.

3.3 Benefits and Limitations

This section delves into where RPA works and where it doesn't. The chapter identifies the key boundaries that need to be kept in mind when in the later chapters the feasibility and value are evaluated stage by stage inside innovation processes.

3.3.1 Benefits

A key benefit for utilizing RPA in business is the increased efficiency and scalability it provides. RPA can execute repetitive tasks much faster than humans and they can operate

around the clock, leading to significant gains in process efficiency. In practice, most organizations adopting RPA report improved productivity and throughput (Artificial Lawyer 2018). Because software "bots" can also be replicated quickly, workload capacity is easily scaled up without a linear increase in people and resources needed. This allows organizations to handle growing volumes of work more flexibly and respond faster to demands.

Since RPA performs tasks with rule-based consistency, it also has drastically reduced error rates compared to manual data entry and processing. Software robots follow prescribed steps exactly each time, eliminating human mistakes such as typos or missed fields. As a result, companies experience higher data quality and fewer processing errors when using RPA. For example, industry analysts note that RPA "eliminates human errors," leading to improved accuracy in data-intensive operations (Artificial Lawyer 2018; Manthaa 2019). This reliability not only minimizes exceptions but also produces more consistent data. Overall, by automating tasks, organizations can achieve near error-free execution of structured processes, improving the accuracy of databases, reports, and transactions.

Another very prominent motivation for RPA is the potential to lower operational costs. Automating high-volume, labor-intensive tasks can yield substantial labor cost reductions and efficiency-related savings. While implementing RPA does need some upfront investment, since it can replace human labor in routine work, run continuously, and scale without major expenses, it often has a favorable return on investment. Many organizations achieve full ROI within months of deployment as work is handled at a fraction of the former cost (Manthaa 2019).

Automating processes executing tasks can easily follow standard operating procedures every time. This reduces the risk of non-compliance caused by human workers skipping steps or taking shortcuts. RPA also produces detailed logs of every action performed, creating an automatic audit trail for each transaction for easy traceability (Deloitte 2025). In sensitive areas like finance or healthcare, consistent adherence to rules and ready auditability is a major advantage of RPA.

One of the key benefits of RPA as evident from the case studies above, is its ability to free up human workers for more high-value tasks. By offloading repetitive tasks to machines, RPA allows human employees to focus on more creative, strategic, and valuable activities. Due to automation, organizations can better utilize their workforce for customer service,

analysis, and inventive projects that a machine cannot do. This not only increases the overall value output of the organization but can also improve employee satisfaction, as workers are relieved from manual tasks.

3.3.2 Limitations

A clear limitation of RPA is its inability to do unstructured and creative tasks. RPA is effective only for well-defined, rule-based processes and struggles with tasks that aren't. Essentially, if a task does not have clear rules or requires intuition, interpretation, or improvisation, basic RPA will fail to handle it (Redwood 2024). In other words, RPA can reliably automate repetitive clicks and data manipulations, but it cannot compose a marketing strategy, design a product, or make executive decisions. Similarly, processes involving a high degree of variability or complex decision logic are ill-suited for plain RPA. In such cases, human expertise or more advanced AI-driven systems are required (Charter Global 2024). This limitation means organizations must carefully select RPA use cases that fit its scope.

Despite the ease of deploying a few bots, scaling RPA across an enterprise and maintaining automated workflows over time can also provide challenges. One issue is that RPA scripts are often tightly tied to the current user interface and process flow. Even minor changes in an application's interface or in process steps can break the operation, requiring reconfiguration (Redwood 2024). Over time, managing a huge number of bots and keeping them updated can also become labor-intensive. As well as this, using RPA to handle hundreds of processes isn't simply a matter of copying bots, since each process may need a custom design. As the RPA program grows, companies often encounter significant maintenance effort and complexity.

Since RPA technology works by mimicking human interactions with software, it requires that the information to be processed is accessible in defined fields or predictable digital documents. If a process involves reading data from unstructured sources, basic RPA tools alone aren't enough. For example, an RPA bot can easily extract an invoice number from the same position in a standardized digital form, but it cannot interpret a poorly formatted email or a handwritten note (Charter Global 2024). As a result, RPA works best as part of a digitized workflow, and organizations must ensure processes are sufficiently standardized and digitized before they can be effectively automated with RPA. An important consideration for

fast-moving innovation environments utilizing RPA can also be the technology's UI-level dependence since upstream systems can change and require the bots to be updated.

As stated above, RPA comes with plenty of technical challenges, but it is also an organizational change and can be met with concern from employees. Workers whose jobs involve the tasks being automated may fear that their roles aren't needed anymore. This anxiety can lead to pushbacks, low morale, or lack of cooperation with RPA programs if not properly managed. Additionally, redeploying staff from automated roles to more analytical or innovative tasks is not easy and requires planning. Thus, the human factor is a significant consideration for successful RPA implementation.

3.4 Role of RPA in Innovation

As established in previous chapters, RPA is most often used as an operational technology that automates high-volume, rule-based, digital tasks by interacting with existing information systems at the user-interface layer. This mode of automation makes RPA fast to deploy since it doesn't really require redevelopment of existing systems. On the contrary, it also makes it constrained to well-structured work. In this sense it is seen that RPA can typically influence innovation indirectly by improving efficiency, quality and compliance in surrounding processes rather than being treated as an element inside innovation management itself. (van der Aalst et al. 2018)

Prior studies on RPA adoption in administrative and transactional functions (e.g., finance, HR, and supply chain) consistently report benefits such as reduced manual effort, faster cycle times, and stronger auditability (Lacity & Willcocks 2016). These effects free employee time and attention for analysis, experimentation and coordination which can support innovation work even when automation is not embedded in the innovation stages themselves. Additionally, research on the digitalization of innovation processes shows a similar pattern. For example, Cooper and Sommer (2016b) describe how digital tools in the Stage-Gate model strengthen innovation by accelerating evidence preparation and review cycles, while leaving creative decision-making to humans.

Although RPA cannot generate novel concepts or make strategic choices, it could also directly support specific activities within innovation processes when those activities are structured or document-heavy for example. Some activities match RPA's strengths (repeatable, rules-based, digital inputs) and can therefore influence cycle time and consistency within the process, not just around it. This is further analyzed in chapter 4 where each innovation framework in the scope of this thesis is dissected in to activities that could or couldn't be automated. (Prucha 2024, van der Aalst et al. 2018)

The effectiveness of RPA depends on task characteristics. It helps when inputs are structured, templates are stable, user interfaces change infrequently and decision rules are explicit. In practice, this means ideation, problem framing and creative synthesis should remain primarily human-led, while documentation, evidence handling, and routine data preparation are examples of stronger candidates for automation. (van der Aalst et al. 2018)

Existing literature discusses in detail what RPA is, where it tends to work, adoption challenges and operating-model choices. However, they do not offer a comparative, stage-level map across established innovation frameworks that identifies where RPA is both feasible and valuable inside those processes. This leaves industry managers without actionable guidance on which stages to automate first and why.

4 Analysis

This chapter applies the rubric from the previous chapter to the established innovation frameworks and reports the stage-level findings. For each framework, the relevant stages are summarized, noting the artefacts and interfaces involved, evaluating the ease and benefit of RPA in them. The results are organized framework by framework, followed by a cross-framework analysis that highlights the highest-priority opportunities. Interpretation is kept to a minimum as the later chapters use these results to explain why the patterns appear and what they imply for both theory and practice.

4.1 Evaluation Criteria

To systematically analyze the suitability of RPA integration within innovation frameworks, a set of clearly defined evaluation criteria must be established. With these criteria, a structured approach for consistent comparison is made possible. The two primary criteria evaluated are ease of implementation as well as perceived benefits. Each stage of the selected innovation frameworks presented in chapter 2 will be evaluated against these two criteria. For practical insights, each criterion will also be given a rating of high, medium or low to indicate applicability for RPA integration.

The first criterion evaluated is perceived benefits which evaluates the potential value and impact achieved by implementing RPA in a specific stage of innovation. For example, if an innovation phase can be seen to gain huge efficiency benefits, it can have a high perceived benefit. Other benefits evaluated include error reduction and improved data quality, as automation can minimize human errors. Additionally, cost reduction, improved compliance in documentation and some indirect benefits such as freeing human workers for other tasks are also considered as factors leading to more perceived benefits.

Ease of implementation in this context is defined as how realistically and practically RPA can be applied to each specific phase of the innovation process. To determine it, plenty of different aspects are considered. The first thing to consider is technical feasibility, this refers to whether the tasks in a specific innovation stage are even suitable for automation based on

RPA capabilities. As identified in chapter 3, ideal tasks for RPA are rule-based, repetitive and involve structured digital inputs. On the contrary, processes requiring significant human judgement or creativity are considered less feasible Additionally, the maturity of the process will also have effect as well defined and standardized processes with minimal variability are easier to automate. As the aim of this thesis is a flexible framework usable across companies and industries, organizational readiness and other organizational factors are excluded from the evaluation.

4.2 Open Innovation

Open Innovation is a framework that promotes external knowledge, potentially enhancing innovation outcomes significantly. To analyze the possibility for RPA integration inside this framework, a deeper dive into its distinct phases is needed. These phases are as identified in chapter 2: idea sourcing, idea integration, idea commercialization and collaboration management.

Idea Sourcing

The idea sourcing phase involves identifying, collecting, and screening ideas from a variety of external sources. This process often generates large amounts of data, making it resource-intensive and prone to manual errors. Integrating RPA in idea sourcing could offer substantial improvements by automating repetitive data-entry tasks, preliminary categorization, and initial filtering based on clearly defined criteria. The consistency of automation also greatly reduces human error, improving data quality and enhancing overall operational efficiency. Moreover, automating these routine tasks could significantly reduce the time required, thereby accelerating initial idea assessment and decision-making processes. This leads to the conclusion that RPA in the idea sourcing phase would have plenty of perceived benefits.

Despite the clear advantages, the implementation of RPA in idea sourcing faces challenges. The primary concern being that the phase can often include sourcing data from diverse formats and platforms leading to unstructured outputs. Since RPA's effectiveness relies on predefined rules, inconsistencies in data formatting or submission methods can substantially reduce automation feasibility. While this can be mitigated by introducing standardization to

open innovation platforms or additional technology like AI, the implementation for this phase would not be easy.

Idea Integration

During the idea integration phase, the ideas sourced in the previous phase are effectively integrated into internal research and development. This phase is heavily reliant on judgment and interpretation, since not all ideas should be developed further. Robotic process automation could streamline some tasks such as compliance checks and basic compatibility evaluations to enhance efficiency and accuracy. However, the main focus of this phase is assessing ideas, which without machine learning or other technological tools still relies heavily on the creativity and analytical knowhow of human personnel. Because these tasks dominate this phase, the perceived benefits are very limited.

As this is the case, robotic process automation in this phase would demand the development of highly structured and predefined criteria for said tasks as well as standardizing the inputs in a structured manner. Consequently, any effective automation would necessitate massive process standardization and potentially additional decision support tools like AI. This complexity significantly hinders the implementation of RPA.

Idea Commercialization

In the commercialization phase, activities such as IP management, regulatory compliance, and data processing related to market testing offer some opportunities for RPA. Tasks like maintaining IP records, tracking compliance documentation, and compiling market data are typically rule-based and repetitive, aligning perfectly with RPA capabilities. Automation in these tasks can significantly reduce manual workload, increase consistency in documentation and compliance activities, and expedite market-entry processes. The operational efficiencies gained here can translate directly into faster commercialization cycles and substantial cost savings. Therefore, perceived benefits are high.

Despite these automation opportunities, commercialization also requires careful consideration of existing infrastructures, data handling practices, and compliance regulations. Ensuring smooth integration of RPA would involve standardizing compliance processes and digitizing legacy documentation practices. Additionally, since commercialization activities frequently involve external partners and diverse regulatory environments, compatibility with

various digital systems could have some practical challenges. These practical complexities lead to difficult implementation.

Collaboration Management

Effective management of collaboration in open innovation demands a lot of coordination, systematic communication, and compliance through agreements. RPA would excel in handling structured administrative tasks such as automated notifications, contract status monitoring, compliance checks, and data reconciliation between stakeholders. Automation in collaboration management could significantly reduce administrative costs and enhance operational efficiency.

While the benefits of RPA in collaboration management are high, variety in stakeholders' technological infrastructures, different levels of digital maturity, and inconsistent compliance standards across partners can complicate automation. Effective RPA integration would require substantial initial setup to harmonize interfaces and communication standards across stakeholders. Real-world execution challenges hinder the ease of implementation, but as standardization in technological infrastructure and compliance rises, implementation of RPA gets easier.

Overall Assessment

Evaluating open innovation through strict criteria reveals moderate potential for RPA integration, particularly in administrative-heavy phases like idea sourcing and collaboration management. The analytical complexity and the flexibility of the framework when it comes to standardization of procedures for example, severely limit RPA application feasibility and ease of implementation. Consequently, RPA adoption within open innovation demands careful selection of processes that align closely with structured and repetitive characteristics, coupled with comprehensive efforts to standardize and digitize process inputs. For further comparison, the utilization of RPA in open innovation is presented in traffic light style in table 1 below.

Table 1: RPA in Open Innovation

	Idea sourcing	Idea Integration	Idea Commercialization	Collaboration Management
Perceived Benefits	High	Low	High	High
Ease of implementation	Low	Low	Low	Medium

4.3 Design Thinking

As stated in chapter 2, Design Thinking in this analysis is represented by the Double Diamond model due to its wide acceptance, clarity, and defined phases. The model emphasizes user-centric innovation through iterative problem-solving. Using the defined criteria, this chapter addresses both the perceived benefits and ease of implementation of RPA in each phase of the double diamond model. These phases are called discover, define, develop and deliver.

Discover Phase

The discover phase is dedicated to understanding user needs and gathering qualitative and quantitative insights through observation and research. RPA's potential in this phase could be automating the data collection processes, especially quantitative data collection and preliminary organization tasks. Automated collection could significantly speed up the analysis of surveys, questionnaires, and other web-sourced data. This could directly reduce manual data processing errors allowing researchers to focus more closely on qualitative insights.

Implementing RPA in the discover phase presents considerable challenges. The data collected during this phase often consists of qualitative insights such as interviews and observations, which are inherently unstructured, making them unsuitable for straightforward automation. RPA requires structured input, meaning that only standardized surveys and quantitative feedback are realistically feasible for automation. Because of this, significant pre-standardization efforts or other technologies would be required, which reduces the ease of implementation.

Define Phase

In the define phase, these insights are made into clear, actionable problem statements. The primary benefit of implementing RPA in this phase would just be in administrative tasks such as categorizing and archiving insights, automating document management, and ensuring consistent documentation practices. The structured documentation done by RPA could enhance accuracy and reduce manual efforts, however these tasks represent relatively minor efficiency improvements in the actual phase itself.

The actual tasks of the phase related to problem definition require considerable human interpretation, creativity and contextual understanding, which are incompatible with RPA. To successfully automate those, would require other technologies like AI and even then, the level of human capabilities might not be reached. The limited structured administrative tasks that could feasibly be automated represent only a fraction of the overall phase activities.

Develop Phase

During the develop phase, teams engage in iterative idea generation and rapid prototyping. While most tasks in this phase also involve creative thinking, there are opportunities for RPA in repetitive supporting tasks, such as maintaining prototype records and tracking iterative changes systematically. Automation here could help streamline workflows and iteration cycles as well as reduce manual errors in administrative processes. In any case, these opportunities are somewhat supportive, leading to low perceived benefits.

Implementing RPA in the develop phase would pose significant practical challenges due to the highly iterative, flexible, and creative nature of prototyping. Since RPA relies on structured and repetitive tasks, integrating it in a dynamic environment would demand extensive structuring and ongoing management of automated systems, presenting practical barriers.

Deliver Phase

In the deliver phase, the developed solutions are refined based on user feedback, which does present some automation possibilities. RPA could be effectively utilized for structured tasks such as collecting user feedback and generating summary reports. Automating these repetitive tasks could significantly increase efficiency and improve responsiveness during the solution delivery process. However, just as in the discover phase, these summaries would be

highly dependent on quantitative data and drawing deliverable features based on the insights would still require a creative mind.

Despite the automation opportunities, practical implementation in the deliver phase could also have challenges. As stated, user feedback often involves qualitative inputs requiring interpretation, limiting RPA's direct applicability without additional AI integrations. Moreover, the phase typically involves rapidly responding to iterative user feedback, which complicates standardization efforts required for effective RPA deployment. Thus, practical feasibility is limited.

Overall Assessment

In summary, the Design Thinking framework, as illustrated by the Double Diamond model, exhibits limited potential for RPA integration due to its inherently creative and iterative nature. Structured tasks suitable for automation exist primarily in administrative and documentation-related activities within certain phases. However, significant implementation challenges consistently limit ease of integration, making substantial RPA deployment realistically feasible only for narrowly defined, standardized tasks. These conclusions lead to a low overall suitability of RPA in design thinking which is illustrated in table 2 below.

Table 2: RPA in Design Thinking

	Discover	Define	Develop	Deliver
Perceived Benefits	Medium	Low	Low	Medium
Ease of implementation	Low	Low	Low	Low

4.4 Lean & Agile Innovation

In this analysis, Lean Startup and Agile Innovation are analyzed collectively, since they overlap majorly in structure and approach, primarily emphasizing iterative processes, rapid experimentation, and continuous customer feedback (Ries, 2011; Rigby, Sutherland & Takeuchi, 2016). Because of this overlap, it is logical to evaluate these frameworks using the Lean Startup's established iterative loop phases, which are build, measure, and learn.

Build Phase

The build phase in Lean Startup involves creating Minimum Viable Products (MVPs) or prototypes to validate hypotheses rapidly. In this phase, RPA could again support primarily

administrative tasks such as maintaining records of development iterations, systematically logging changes and automating other routine activities. Automating these repetitive tasks could enhance efficiency, accuracy, and compliance consistency, offering modest yet clear operational improvements.

However, RPA's practical implementation in the Build phase faces barriers. The process typically involves rapidly changing tasks, frequent adjustments based on creative insights, and prototypes often varying significantly in complexity and structure. Standardizing these tasks sufficiently to implement effective automation would require substantial upfront structuring and continuous system management. The dynamic, flexible nature of building MVPs thus severely restricts the straightforward implementation of RPA.

Measure Phase

The measure phase, which focuses on gathering and analyzing user feedback and performance data, presents clearer opportunities for RPA integration. Automation here can significantly streamline tasks such as structured data collection from customer interactions, automated survey processing, customer usage pattern analysis, and automatic compilation of feedback reports. For example, automating the extraction of structured customer data from various platforms would enhance data accuracy and expedite decision-making cycles. Due to the structured nature and repetitive tasks within this phase, the perceived benefits are high.

Despite clear benefits, some implementation challenges remain. Customer feedback, although structured in certain instances, frequently includes qualitative elements requiring interpretation, complicating the automation process. However, the necessary standardization and data format consistency for RPA implementation can be achieved fairly simply by using well established survey tools for example.

Learn Phase

In the learn phase, the measured results are analyzed. Automating routine data summarization and tracking strategic adjustments can offer benefits in terms of operational efficiency and structured decision documentation. For instance, using RPA to systematically record insights can provide enhanced transparency in decision-making processes. Nonetheless, these benefits are just supportive, thus lowering the perceived benefits.

Ease of implementation in the learn phase is highly restricted due to the intrinsically analytical and strategic nature of tasks involved. The interpretation of user feedback and decision-making processes are fundamentally dependent on human intuition, creativity and understanding of market dynamics. Automating these elements exceeds RPA's current capabilities significantly. Consequently, this phase has considerable implementation challenges, resulting in exceptionally low ease of implementation.

Overall Assessment

Lean Startup and Agile Innovation methodologies present a limited but distinct set of opportunities for RPA integration, primarily in structured administrative, documentation, and data processing tasks. However, significant practical challenges due to the highly iterative, flexible, and creative nature of these frameworks limit substantial automation. Effective RPA application demands rigorous task standardization, structured data management practices, and potentially complementary AI-driven tools. Table 3 below reflects these findings.

Table 3: RPA in Lean & Agile Innovation

	Build	Measure	Learn
Perceived Benefits	Medium	High	Low
Ease of implementation	Low	High	Low

4.5 TRIZ

TRIZ (Theory of Inventive Problem Solving) is evaluated in this study as a structured innovation approach, providing systematic tools for inventive solutions through clearly defined phases: problem definition, contradiction identification, abstraction to a higher-level problem, solution identification, and solution refinement (Altshuller, 1984; Gadd, 2011). Given TRIZ's emphasis on structured problem-solving tools, each phase's evaluation specifically addresses potential RPA integration with these tools.

Problem Definition Phase

This phase involves clearly defining technical or functional problems, typically utilizing structured documentation and systematic information management. RPA can automate routine documentation, ensuring structured categorization, consistency checks, and version

control. Particularly beneficial would be automating standard data entry, maintaining systematic records of problem definitions, and tracking historical issue logs. These automation tasks enhance operational accuracy and efficiency, reducing administrative overhead and errors.

Implementation of RPA in problem definition can be practically feasible, as the phase relies significantly on structured data and clearly defined technical documentation. However, this assumes a high degree of standardization of the documentation processes, which might vary significantly in different technical fields or organizations. Achieving the necessary consistency in technical documentation and terminology for effective automation may present initial hurdles.

Contradiction Identification Phase

Identifying contradictions involves structured analysis of technical parameters, typically employing TRIZ tools like the Contradiction Matrix. RPA could significantly enhance efficiency in automating data retrieval and initial screening processes from technical databases, systematically highlighting patterns, and preparing structured input data for the Contradiction Matrix. This structured support would considerably streamline the initial analysis and ensure data consistency.

While the Contradiction Matrix itself offers structured procedures suitable for automation, practical RPA implementation faces complexities related to consistent data formatting across databases. Technical contradictions often require judgment regarding parameter selection, potentially limiting straightforward RPA integration without extensive standardization or supplementary decision-making tools. Thus, realistic implementation can be difficult.

Abstraction to Higher-Level Problem Phase

Abstraction involves transforming specific problems into generalized forms using structured TRIZ methods, such as substance-field analysis and functional analysis. Automation could support systematic data handling, routine abstraction documentation, and standard record-keeping. Automating these repetitive processes would enhance documentation consistency, administrative efficiency, and reduce manual error. Nonetheless, the benefit remains moderate due to the inherently analytical and interpretative nature of abstraction.

Despite structured elements, abstraction processes in TRIZ significantly depend on human analytical capabilities and deep technical understanding. Automating nuanced abstraction steps would require advanced interpretation capabilities beyond standard RPA. Consequently, automation would be limited to administrative support and structured documentation management tasks. These limitations markedly reduce ease of implementation.

Solution Identification Phase

In the solution identification phase, TRIZ tools such as the 40 Inventive Principles and Trends of Technological Evolution guide systematic solution generation. RPA could efficiently manage structured documentation of potential solutions, systematically tracking generated solutions, updating databases, and performing automated categorization tasks. Additionally, automated searches in patent databases or historical solution archives could streamline initial solution screenings significantly. These structured automation tasks provide clear operational improvements, hence perceived benefits are extremely high.

Although automation of administrative and preliminary screening tasks is feasible, identifying solutions involves substantial interpretative and creative processes, usually requiring human intervention. RPA alone cannot replicate the complex cognitive tasks involved in creatively applying principles or interpreting patent literature. Therefore, ease of implementation is limited by these challenges.

Solution Refinement Phase

Refinement involves detailed assessment of the feasibility of solutions, involving systematic evaluation and iterative improvements. RPA could streamline administrative documentation, maintaining iterative solution records, compliance checks, and systematically managing structured feasibility criteria documentation. Such automation would ensure documentation consistency, reduce manual error and streamline administrative workflow, representing modest benefits.

Solution refinement is inherently iterative and analytical, heavily relying on human judgment and expert evaluations of feasibility. Automating these sophisticated evaluations with RPA is impractical, as it demands complex judgment and analysis. Only the administrative and structured documentation aspects present feasible automation opportunities, significantly restricting ease of implementation.

Overall Assessment

TRIZ, being structured and systematic, provides moderate yet specific opportunities for RPA integration predominantly in administrative, documentation-related, and preliminary screening tasks. Each phase's practical implementation faces considerable limitations due to the high level of human interpretation and analytical complexity inherent in TRIZ methods. The summarization in table 4 below clearly reflects these considerations.

Table 4: RPA in TRIZ

	Problem Definition	Contradiction Identification	Abstraction to Higher-Level Problem	Solution Identification	Solution Refinement
Perceived Benefits	Medium	High	Medium	High	Medium
Ease of implementation	Medium	Low	Low	Medium	Low

4.6 Stage-Gate

Stage-Gate is a highly structured and systematic innovation framework, providing clearly defined phases for developing and launching new products. Due to its structured nature, it inherently aligns with the rule-based and repetitive capabilities of Robotic Process Automation (RPA). This evaluation addresses potential RPA integration specifically within each distinct Stage-Gate phase: Scoping, Business Case Development, Development, Testing/Validation, and Launch (Cooper, 2008; Cooper & Sommer, 2016a).

Scoping Phase

During the scoping phase, preliminary assessment tasks such as market scans, initial competitor analyses, and resource availability checks are frequent. RPA could offer significant benefits by automating structured data collection from market databases and systematically documenting initial assessments. For example, automated market data aggregation and preliminary financial calculations can improve speed, accuracy, and consistency.

RPA implementation in the scoping phase also faces moderate challenges. While the tasks are often straightforward and suitable for automation, variability in data formats across external sources may require considerable upfront standardization or integration efforts. Thus, implementation is moderately difficult.

Business Case Development Phase

In the business case development phase, RPA could significantly enhance operational efficiency through automation of structured financial modeling, routine scenario analysis and systematic documentation management. Automated tasks would include extracting standardized financial data and performing structured cost-benefit analyses. These structured, rule-based activities provide substantial benefits, improving speed, accuracy, and reducing manual errors.

Although structurally feasible, this phase's automation implementation demands precise standardization and consistency in financial modeling inputs. Given the complex nature of financial and strategic data from varied sources, ensuring consistent automation can pose considerable initial implementation challenges.

Development Phase

The Development phase typically involves detailed product design and initial prototyping. Automation opportunities here again lie primarily in administrative and routine documentation tasks, such as maintaining project documentation and managing structured compliance checks. Although these tasks are more supportive than central to innovation itself, they provide clear efficiency and consistency improvements.

Implementation of RPA in the Development phase encounters notable complexity due to iterative and creative design tasks that inherently require significant human intervention and decision-making. Automation would be confined to routine administrative and structured documentation activities, limiting overall applicability.

Testing/Validation Phase

During the testing/validation phase, structured automation opportunities are significant. Tasks suitable for RPA include automating data collection from standardized testing environments, systematically compiling test results, managing structured documentation and performing rule-based compliance checks. For example, automating validation data

processing and generating standardized reports can greatly enhance speed, consistency and accuracy. Therefore, perceived benefits in this phase are notably high.

Practical RPA implementation is relatively straightforward here due to highly structured and repetitive validation activities. Standardized testing protocols, systematic data formats, and routine documentation procedures significantly facilitate automation. Minimal complexity and clear rules make this phase particularly feasible for RPA implementation.

Launch Phase

In the launch phase, structured tasks include systematic preparation of launch documentation, structured tracking of market entry activities, compliance checks, and post-launch monitoring data management. Automating these structured tasks offers tangible improvements in operational efficiency, data consistency and reduces administrative workload. However, the actual strategic and creative elements of launching products significantly limit RPA benefits, hence the perceived benefits are moderate.

Automation in the launch phase faces moderate implementation challenges due to variability in market conditions, diverse external stakeholder engagements and regulatory variations across different market entries. Standardizing these interactions sufficiently to allow automation can be complex, limiting ease of implementation.

Overall Assessment

The Stage-Gate model offers significant and clear potential for RPA integration, especially in structured, rule-based tasks. However, phases involving iterative, creative, or strategic decision-making activities inherently limit automation feasibility. Effective RPA implementation within Stage-Gate requires upfront standardization, data structuring, and possibly integration with complementary analytical tools. These insights are clearly reflected in table 5 below.

Table 5: RPA in Stage-Gate

	Scoping	Business case development	Development	Testing / Validation	Launch
Perceived Benefits	High	High	Medium	High	Medium
Ease of implementation	Medium	Medium	Low	High	Medium

4.7 Cross-Framework Comparison

The overall results of the analysis seem to follow the working assumption of more structured frameworks providing clearer opportunities for automation, whereas creative and judgment-heavy phases remain largely human-led. Across all five frameworks, the strongest opportunities concentrate where inputs and outputs are standardized, templates exist, and documentation or is central to the phase. This is most visible in Stage-Gate as well as to some extent in TRIZ. In contrast, phases that rely on interpretation and rapid change show only limited potential beyond administrative support. This applies especially to Design Thinking and to judgment-centric parts of Open Innovation.

The Testing/Validation phase in the Stage-Gate model stands out as the clearest fit for robotic process automation. Activities such as collecting test results, compiling gate material, and running compliance checks are repetitive and rule based. The work runs on stable templates and structured data, which makes automation straightforward and beneficial for speed, accuracy and traceability. The Measure phase in the Agile Innovation framework is another strong focus. Structured data collection, survey handling, log consolidation and dashboard refreshes can be automated without disturbing the iterative loop. Automation here improves decision speed and documentation quality between cycles. Administrative and search-oriented activities in TRIZ (e.g., patent database searches) align with RPA's strengths. Collaboration management and Commercialization in Open Innovation also include recurring tracking, IP registers and other coordination tasks. These can benefit from automation, but diversity in systems often reduces feasibility until interfaces and formats are harmonized.

Design Thinking is dominated by qualitative inquiry, framing and creative iteration. In its phases, RPA is mostly limited to narrow administrative support (e.g., file handling, simple feedback collection). Idea generation phases in all frameworks usually rely on expert judgment to select and adapt external ideas hence there being little direct role for RPA beyond documentation. Two factors can be seen across frameworks. First, task structure and input format dominate outcomes, since structure provides direct value paths for RPA while phases centered on creative synthesis do not. Second, interface stability and external heterogeneity matter. Internal, stable systems and templates raise feasibility, while multi-partner environments and changing UIs increase maintenance and exceptions.

5 Discussion

In this chapter the results reported in the previous chapter are interpreted to answer the study's research questions. The focus is explaining observed patterns and their implications for both theory and especially practice. The discussion is structured directly around the research questions. Firstly, the sub research questions: "How easily can RPA be implemented to each stage of the established innovation process frameworks?" and "What measurable benefits does RPA bring to each stage of the established innovation process frameworks?" are answered by interpreting what drives implementation ease as well as where benefits concentrate. These perspectives are later synthesized to answer the main research question: "Where, within established innovation process frameworks, can Robotic Process Automation (RPA) deliver measurable value?" After interpreting the analysis results, the theoretical and managerial implications are discussed to conclude with a practical roadmap for RPA implementation.

5.1 Ease of RPA Implementation in Innovation

For the first sub research question: "How easily can RPA be implemented to each stage of the established innovation process frameworks?", the results point to a simple organizing principle. Implementation ease follows the structure of work. RPA can be integrated with few surprises when the stages revolve around predictable and explicit rules. On the other hand, when there is a lot of fluid parts, collaboration and interpretation involved, the ease falls away.

Across frameworks the same pattern recurs. Stages that compile evidence, check completeness, normalize formats and shuttle information between systems are straightforward because they expose deterministic steps over stable inputs. Stage-Gate therefore shows high ease in testing/validation where deliverables are templated and checks are routine. Lean/Agile is favorable around measurement and reporting. Open Innovation is mixed, intake and compliance can be channeled through standard forms, but heterogeneity in partner tools reduces interface stability. TRIZ divides cleanly between routinised retrieval and screening, which are automatable, and its core inventive moves, which are not. Design Thinking sits at

the difficult end in discovery/definition/ideation because inputs are qualitative and outcomes are synthesized rather than computed, although research-operations activities around these stages can be automated.

Four interacting conditions explain the observed ease scores. Deterministic vs interpretive tasks. Machine-readable inputs, such as spreadsheets with constrained fields, CSV/JSON or structured PDFs, keep variance low while free text introduces difficulties. Stable interfaces limit maintenance while volatile UIs and unsynchronized partner portals do the opposite. Finally, process governance (versioned templates, canonical identifiers, documented exception paths etc.) reduces ambiguity. These mentioned conditions can compound, meaning a deterministic task can still be hard if inputs are messy or the UI shifts weekly, while modest pre-standardization can put borderline candidates into the “easy” range.

Two kinds of exceptions are noteworthy. Firstly, some creative stages conceal administrative sub-tasks such as participant scheduling, consent handling, survey export and repository hygiene that are both necessary and automatable, even when the surrounding work remains human-led. Secondly, boundary conditions can flip ease profiles over time. Stabilized interfaces immediately improve durability. Simple digitization like mandating a single gate dossier template can move a stage from low to moderate ease without changing any of the tools.

The practical implication here is that ease is, to a significant degree, designable. Organizations can raise it by unifying templates and identifiers, replacing ad-hoc email attachments with controlled submissions and negotiating API access with vendors and partners. They lower it when they tolerate parallel “equivalent” templates or anchor production-critical steps to volatile prototype tools.

With this in mind, the framework-specific outcomes are unsurprising. In summary, RPA is easiest to implement where tasks are deterministic, inputs are structured and interfaces are stable under basic process governance. Many “hard” stages can be made easier through lightweight standardization and digitization.

5.2 Measurable Benefits of RPA in Innovation

In this chapter the focus of the analysis is not feasibility but payoff as the second sub research question asks: “What measurable benefits does RPA bring to each stage of the established

innovation process frameworks?" The findings revolve around four metrics: cycle time, accuracy/quality, compliance/auditability and cost. Gains rise when bots remove queuing and rework in routine evidence handling, apply rules consistently and produce standardized artefacts with complete activity logs. Cost effects follow as a mix of reduced manual effort and avoided downstream waste.

Benefits are largest where stages already produce well-formed artefacts and accept automation-generated outputs as authoritative. In Stage-Gate's testing/validation, RPA shortens the time to assemble and verify decision packs, lowers cross-document inconsistency and creates auditable trails. These are all tangible improvements visible in reduced lead time to gate and fewer rework cycles. Business-case work also improves when input sources are known, with bots pre-populating templates and performing "sanity checks". TRIZ benefits come in retrieval, triage and evidence packaging which can reduce omission errors without touching invention.

In Lean/Agile, payoff concentrates in the measure part. Automating data collection, normalization and distribution compress the loop between build and insight without altering methods. Accuracy improves through routine validation, which reduces false learning and the share of sprint capacity spent fixing instrumentation. The build and learn phases benefit mainly at their administrative edges, where evidence packaging and cross-tool synchronization can be measured. Open Innovation shows a divisive distribution of benefits. When intake and compliance flow through controlled mechanisms, bots can lower onboarding times, reduce missing-field and expired-document rates and improve audit readiness. Co-creation and negotiation, by contrast, yield little measurable gain because success depends on human alignment rather than throughput. In Design Thinking, benefits are modest in qualitative cores but clearer in research operations, where consent tracking, survey export and repository hygiene reduce administrative overhead and improve completeness. These metrics are easy to observe.

Realizing measurable gains from RPA depends on several straightforward preconditions. Credible evaluation requires established baselines and improvements should be evidenced through before–after comparisons against set targets. Acceptance by downstream actors is equally important also important. If auditors do not recognize bot-generated artefacts as official, parallel manual routines persist and dilute any benefits. Data hygiene and interface

stability also condition outcomes, as exceptions and frequent UI changes consume the very capacity that automation is intended to release.

The results also indicate ceilings and diminishing returns. Once obvious defects are removed, additional checks yield small marginal benefits and may lengthen lead time. In mature analytics environments, automating another step-in dashboard refresh typically contributes less than improving experimental design. These patterns motivate periodic re-prioritization after the initial wave of automations, shifting attention to the next constraint, which could be reliability or cross-system synchronization rather than local throughput.

Finally, the evidence supports a clear boundary. Benefits are lost when RPA is extended into the interpretive core of stages where judgement, synthesis or negotiation generate value. The productive configuration should be to surround those cores with automation so that human effort concentrates on analysis and design. Framed in this way, the outcomes are consistent: shorter and more reliable decision lead times, fewer avoidable defects, stronger audit posture at lower cost and a visible reallocation of effort from rote handling to activities that advance innovation.

In sum, RPA delivers its most reliable, measurable gains in artefact-heavy, rule-governed stages, with the largest effect sizes in cycle time, accuracy and auditability and cost improvements emerging as those primary gains compound. These benefits are also consistent with what prior RPA research have reported in other domains (Lacity & Willcocks 2016).

5.3 Value of RPA in Innovation

This section integrates the feasibility and payoff perspectives to answer the study's main research question: "Where, within established innovation process frameworks, can Robotic Process Automation (RPA) deliver measurable value?" The synthesis treats implementation ease and benefits as two coordinates on a single map. The relevant opportunities are not evenly spread but rather they cluster in recognizable zones that cut across frameworks because they are defined by the work's structure rather than the process definitions.

The first and most reliable zone is where deliverables are well-formed and decision logic is explicit. Here, ease is high and benefits are large because automation can replace queuing and rework without touching judgement. In practical terms, this zone includes the evidence-

handling part of Stage-Gate (testing/validation) and the measure part of Lean/Agile. In both cases, inputs are already constrained, outputs are templated and the consequences of small errors are visible to reviewers. RPA can therefore compress cycle time, raise data quality and strengthen audit trails with minimal fragility. These stages form the core sweet spots for value because feasibility and payoff reinforce one another rather than trading off.

A second zone comprises stages where benefits are meaningful, but ease depends on local conditions. The business-case workstream in Stage-Gate, intake and compliance administration in Open Innovation and contradictions in TRIZ belong here. Each offers measurable improvements such as faster assembly of scenarios, fewer missing fields and expired documents as well as quicker and more complete screening, yet each is sensitive to the degree of standardization in inputs and to the stability of the interfaces involved. In these contexts, small design choices (a single canonical template, controlled vocabularies, a stable submission front door, or access to an API) can make opportunities reliably valuable. They are conditional sweet spots as the value is there, but it is unlocked only once the surrounding process is made legible.

A third zone contains activities that are easy enough to automate but whose contribution to innovation outcomes is modest. These are housekeeping tasks around otherwise human-centered work such as filing, renaming, status updates etc. They rarely change decision latency or error rates at scale. While worth capturing to reduce nuisance work, they should not displace higher-yield targets. They are useful, low risk, but unlikely to register as material improvements in the metrics used for this thesis.

At the opposite extreme are stages where the potential benefit might appear high, but ease remains stubbornly low because the value-adding moves are interpretive. Early discovery and ideation in Design Thinking, co-creative collaboration in Open Innovation and the core inventive steps of TRIZ fall into this category. Attempts to automate the core of these stages tend to introduce rigidity and exception handling that erase not only feasibility but also net benefit. In these domains, RPA delivers minimal value ensuring clean intake, complete records and reproducible evidence around human work, rather than substituting for it.

Seen as a whole, the synthesis suggests a simple sequencing rule. Automation should begin with the core sweet spots, Stage-Gate evidence handling (testing/validation) and Lean/Agile measurement. Then expanding into the conditional sweet spots once pre-standardization

closes the gap between messy reality and automation needs. Additional gains in these parts are real but rely on a tighter operating envelope. After that refraining from automating the interpretive cores and instead surrounding them with reliable administrative rails so that human attention is conserved for analysis, design and negotiation is suggested. Capturing maintenance wins opportunistically to remove irritants can also be done, but it is crucial to avoid confusing their convenience with strategic impact.

Opportunities for RPA change over time because ease and benefit are not fixed. Small design or process changes can move a stage into or out of the high-value zone. Standardizing a gate dossier, stabilizing a submission interface, or controlling a frequently changing form can make automation both easier and more beneficial. For this reason, priorities should be reviewed regularly as the best candidates in the first wave may not be the same later. The main sweet spots are stable even if the weighting between ease and benefit changes. Stage-Gate's testing/validation and Lean/Agile's measure remain top-ranked because both dimensions are strong. Conditional stages move up when standardization improves, small maintenance tasks rarely rise in priority, and interpretive stages seldom qualify unless the work is redesigned.

The answer to the main research question is therefore more of a work profile and not a framework list. RPA delivers the most value where tasks are rule-based, outputs standardized, and interfaces stable. This finding is consistent with the broader research showing digitalization as an accelerant for innovation (Cooper & Sommer 2016b). Opting for more structured innovation processes allows more opportunities for automation but pockets can also exist in less structured settings if standardization is introduced first. Conversely, stages driven by judgement or creativity even in structured innovation processes gain little beyond supportive administration. In short, the best results come from targeting the predictable, artefact-heavy parts of innovation while leaving the interpretive core to people.

5.4 Theoretical Implications of the Results

This thesis places RPA inside innovation work, but not at its creative core. The results show a division of two layers in each framework. First being a practical, routine layer (collecting evidence, preparing gate packs, keeping records, pulling metrics), where RPA fits very well. The second creative layer (sense-making, ideation, negotiation) can also see some help from RPA, but it does not replace the human aspect in it.

The key idea noticed is that automatability depends less on the framework and more on how clear and structured the work in a specific stage is. If tasks have clear rules, inputs are in standard formats, and the tools don't change all the time, RPA is easier and more useful. Importantly, these are design choices that teams integrating RPA can consider. Also, as companies integrate multiple different frameworks when it comes to their innovation approach, it becomes crucial to identify the common factors that easily automatable stages share. Where work is clear and structured and the organization accepts bot-produced outputs as official, RPA creates measurable value. RPA does not make innovation more rigid by itself, it reduces noise and delays in the administrative parts of the process, which frees time and attention for better experiments and stronger analysis.

Another key implication is that the opportunity map is not fixed. For example, when a partner portal stabilizes in open innovation, a previously fragile idea can become a good automation target. When tools or logins change constantly, the same automation goal can become difficult. In other words, both the ease and the benefits move with the environment, so priorities should be reviewed from time to time.

For future work, it would be useful to define simple measures for how legible a stage is, meaning clear rules, data structure and interface stability for example. Additionally, the quality of the stated "administrative scaffolding" and how much expert time could be shifted from these routine tasks to creative analysis needs to be quantifiable to make before/after comparisons clearer. Additionally, the results of this thesis could also be connected to established innovation theory like research on organizational ambidexterity for example. In that context, RPA could be seen as an enabler of ambidexterity as it strengthens the exploitation side by making routine evidence-handling and compliance more consistent and less costly. At the same time, this would free resources for exploration, allowing teams to focus on creativity without being burdened by administration.

5.5 Validity, Limitations and Transferability of the Results

This study is based on a structured literature review and a rubric-driven interpretation of where RPA fits inside established innovation frameworks. As such, several validity questions arise. First, there is judgement in scoring. Even with explicit criteria, ease and benefit assessments rely on the researcher's interpretation of task structure, input quality, and

interface stability. To reduce bias, the rubric was kept simple and applied consistently. However, different analysts might weigh borderline cases differently. Second, source quality varies. Peer-reviewed papers and foundational books sit alongside industry white papers and vendor case studies. To mitigate this, claims were triangulated across multiple sources where possible and the analysis prioritized patterns that recur across contexts rather than single case studies of dramatic improvements. A third limitation identified is the measurement baselines. Many reported benefits in the literature are before/after comparisons without strict experimental control. This thesis therefore treats benefit claims as credible when linked to clear metrics (lead time, error rates, missing-document rates) and interprets large effect sizes cautiously.

Internal validity is helped by using the same ease & benefit frame across all innovation frameworks. However, there is still a risk to construct validity because ease here is an indicator of how maintainable an automation is and benefits combine several different outcomes into one score. Testing different weightings for these factors would show us that the top opportunities stay at the top, but the middle-ranked stages can shift if the assumptions are changed.

Regarding external validity and transferability, the patterns should travel best to organizations with some process standardization, digitally accessible artefacts and relatively stable tooling. Highly regulated industries often meet these conditions and may see stronger, more measurable benefits. More fluid ecosystems such as start-up companies will likely need some pre-standardization to unlock similar gains. Small firms can still apply the logic, but their opportunities may concentrate on a few high-leverage workflows rather than across an entire framework. Finally, as interfaces, governance and data practices evolve, the opportunity map can fluctuate. Replication with new data, or in a single organization using detailed process logs, would strengthen the claims.

5.6 Implementation Roadmap

The findings of this thesis suggest a simple starting point for organizations considering RPA in innovation management. The main role of automation is to support the routine and documentation-heavy parts of innovation so that people can focus on analysis, design, and

decision-making. The roadmap for implementation is presented in figure 4 below and the later chapter outlines how these insights can be best applied in practice.

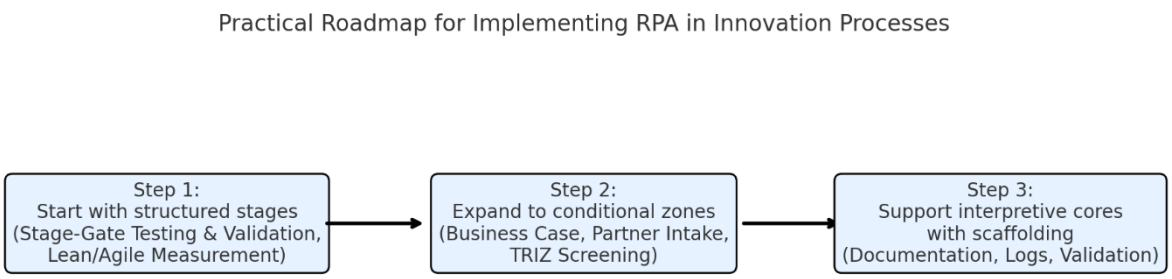


Figure 4: Practical Roadmap for Implementing RPA in Innovation Processes

The first and most obvious step would be to target stages where both feasibility and benefit are high. Starting in these stages identified in the analysis reduces implementation risk while creating visible gains in speed, accuracy, and auditability. Before implementation, however, a degree of pre-standardization is necessary. Agreeing on unified templates and submission channels, for example, can reduce variation and make automation easier to maintain. At the same time, clear measurement baselines should also be defined so that results can be evaluated and communicated.

A recommended approach could be to begin with a narrow pilot, ideally owned by a clearly identified team. The scope should remain small, for example automating the compilation of test results or the generation of a recurring measurement report. Running the bot in parallel for a short period can build trust and ensure that outputs are accepted by reviewers. Once confidence is established, the manual path can be retired.

Scaling should then follow patterns rather than individual tools. Once templates, folder structures, and validation rules are in place, they can be reused in adjacent stages such as business-case preparation or intake processes in Open Innovation. However, for every extension it should always be considered if the inputs remain messy or interfaces unstable. If so, automation should wait until the process is stabilized. Each automation should also have a named owner, documented exceptions, and clear rules for when human review is required. Regular reviews of metrics and maintenance effort ensure that the system continues to deliver value.

Risks such as over-automation of judgement tasks or silent errors can be managed by keeping automation close to routine documentation and away from interpretive cores. By following this path, organizations can capture measurable improvements while keeping the creative and strategic parts of innovation firmly in human hands.

6 Conclusions and Future Research

This thesis set out to answer the main research question: “Where, within established innovation process frameworks, can Robotic Process Automation (RPA) deliver measurable value?” The analysis shows that RPA is best suited to the structured and artefact-heavy parts of innovation processes. Stages such as Testing and Validation in Stage-Gate and Measurement in Lean/Agile consistently emerge as high-value opportunities because they combine well-defined deliverables with clear rules and fairly stable interfaces. Other stages can also produce measurable value once pre-standardization and appropriate governance is in place. By contrast, stages centered on interpretation and creativity were less suitable for automation excluding surrounding administrative tasks.

The main contribution of this work ended up showing that the question is less about individual frameworks themselves and more about the profile of work inside them. RPA delivers value where tasks are repeatable and well structured, regardless of which framework stage they belong to. This perspective shows that innovation management should be viewed as a mix of creative, human-led work and routinized “scaffolding” that can be supported and even replaced through automation.

While these findings do answer the research questions, they are also shaped by the methods used in this study. The scoring of ease and benefit relies on literature and documented cases rather than on live organizational data. Because of this, the most important direction here is future research and practice. Empirical studies are needed to test the implementation roadmap in real organizations. Data on cycle times, error rates, or audit outcomes before and after RPA pilots would provide a stronger evidence base than literature synthesis alone. Comparative studies across industries could also clarify the rate that identified conditions such as regulation, data maturity or process stability make automation easier or harder to implement.

At the same time, practice can benefit from using this thesis as a guide for incremental adoption. As stated, organizations should begin with narrow pilots in structured stages, build trust in the outputs, and then extend the approach by reusing templates and rules. Periodic reprioritization is also essential as an activity may become more or less feasible as interfaces

change for example. This dynamic view also opens a space for combining RPA with other tools in the future. For example, process mining could help identify high-volume candidates, while low-code or API-based automation could take over where UI-level bots struggle.

Another possibility for future research is to look at the interaction between automation and creativity. This thesis has argued that RPA should not replace interpretive work, but there is still value in studying how stronger administrative scaffolding might indirectly improve ideation or problem-solving. If RPA consistently reduces delays and rework, it could allow teams to experiment more frequently or spend more time in synthesis without being burdened by documentation tasks. Measuring these indirect effects already identified in literature, would extend the understanding of how automation influences innovation outcomes more broadly.

As a final takeaway, it is essential that organizations should treat automation as an ongoing capability rather than a one-time project. Templates, vocabulary, governance, and metrics should evolve with the process, ensuring that automation continues to deliver measurable improvements. Academic research can and should support this by developing clearer constructions to make it easier to compare results across contexts. In sum, this thesis has shown that RPA can contribute meaningfully to innovation management, but only in the right places and under the right conditions. The most immediate opportunities lie in the structured, documentation-heavy parts of established frameworks, while longer-term gains depend on standardization, governance, and careful integration with human-led work. Future studies that move from literature to practice will be key to refining this roadmap and demonstrating its impact on both efficiency and innovation outcomes.

References

Altshuller, G. 1984. Creativity as an Exact Science: The Theory of the Solution of Inventive Problems. Gordon and Breach.

Altshuller, G. 1999. The Innovation Algorithm: TRIZ, Systematic Innovation and Technical Creativity. Technical Innovation Center.

Antonsk. 2021. Artefacts and actions in a lean startup cycle. [Image]. Available at: https://commons.wikimedia.org/wiki/File:Lean-StartUp-Cycle-EN.jpg [Accessed 29 May 2025].

Artificial Lawyer. 2018. Two Thirds of Large Corporates Implementing RPA – Deloitte Report. Available at: https://www.artificiallawyer.com/2018/10/11/two-thirds-of-large-corporates-implementing-rpa-deloitte-report/ [Accessed 29 May 2025].

Baker Tilly. 2023. Five ways RPA empowers employee innovation. Available at: https://www.bakertilly.com/insights/five-ways-rpa-empowers-employee-innovation [Accessed 29 May 2025].

Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R. C., Mellor, S., Schwaber, K., Sutherland, J., & Thomas, D. 2001. Manifesto for Agile Software Development. Available at: https://agilemanifesto.org/ [Accessed 29 May 2025].

Blank, S. 2013. Why the Lean Start-Up Changes Everything. Harvard Business Review. Vol. 91, No. 5, pp. 63–72. Available at: https://hbr.org/2013/05/why-the-lean-start-up-changes-everything [Accessed 29 May 2025].

Blue Prism. 2025a. How Jaguar Land Rover Stays Ahead with Smart Automation. Available at: https://www.blueprism.com/resources/case-studies/how-jaguar-land-rover-stays-ahead-with-smart-automation/ [Accessed 29 May 2025].

Blue Prism. 2025b. Santander Reduces Time to Onboard Newcomers by 85% with Blue Prism. Available at: https://www.blueprism.com/resources/case-studies/santander-reduces-time-to-onboard-newcomers-by-85-with-blue-prism/ [Accessed 29 May 2025].

Blue Prism. 2025c. Zurich UK Invests in Customers and Employees with a Successful Intelligent Automation Program. Available at: https://www.blueprism.com/resources/case-studies/zurich-uk-invests-in-customers-and-employees-with-a-successful-intelligent-automation-program/ [Accessed 29 May 2025].

Brown, T. 2008. Design thinking. Harvard Business Review. Vol. 86, no. 6, pp. 84–92. Available at: https://hbr.org/2008/06/design-thinking [Accessed 29 May 2025].

Cavallucci, D. & Rousselot, F. 2011. From TRIZ to OTSM-TRIZ: Addressing complexity in inventive design. Procedia Engineering. Vol. 9, pp. 86–91. Available at: https://doi.org/10.1016/j.proeng.2011.03.103 [Accessed 29 May 2025].

Charter Global. 2024. RPA, Automation, and Intelligent Automation: Choosing the Right Approach. Available at: https://www.charterglobal.com/comparison-between-rpa-automation-and-intelligent-automation [Accessed 29 May 2025].

Chesbrough, H. W. 2003. *Open Innovation: The New Imperative for Creating and Profiting from Technology*. Boston, MA: Harvard Business School Press.

Cooper, R. G. 1990. Stage-Gate Systems: A New Tool for Managing New Products. Business Horizons. Vol. 33, No. 3, pp. 44–54. Available at: https://doi.org/10.1016/0007-6813(90)90040-I [Accessed 2 October 2025].

Cooper, R. G. 2008. Perspective: The Stage-Gate Idea-to-Launch Process—Update, What's New and NexGen Systems. Journal of Product Innovation Management. Vol. 25, No. 3, pp. 213–232. Available at: https://doi.org/10.1111/j.1540-5885.2008.00296.x [Accessed 2 October 2025].

Cooper, R. G. 2011. Winning at New Products: Creating Value Through Innovation (4th ed.).

Cooper, R. G. & Sommer, A. F. 2016a. Agile–Stage-Gate: New Idea-to-Launch Method for Manufactured Products is Faster, More Responsive. Industrial Marketing Management. Vol. 59, pp. 167–180. Available at: https://doi.org/10.1016/j.indmarman.2016.10.006 [Accessed 2 October 2025].

Cooper, R. G. & Sommer, A. F. 2016b. The Agile–Stage-Gate Hybrid Model: A Promising New Approach and a New Research Opportunity. Journal of Product Innovation

Management. Vol 33, No. 5, pp. 513–526. Available at: https://doi.org/10.1111/jpim.12314 [Accessed 2 October 2025].

Creswell, J. W. & Creswell J. D. 2017. Research Design: Qualitative, Quantitative and Mixed Methods Approaches (5th ed.).

Deloitte. 2025. Financial Reporting, RPA, Risks and Controls. Available at: https://www2.deloitte.com/us/en/pages/audit/articles/financial-reporting-rpa-risks-and-controls.html [Accessed 29 May 2025].

Design Council. 2004. The Double Diamond: A universally accepted depiction of the design process. Available at: https://www.designcouncil.org.uk/our-resources/the-double-diamond/ [Accessed 29 May 2025].

Gadd, K. 2011. TRIZ for Engineers: Enabling Inventive Problem Solving. Wiley. Available at: https://www.wiley.com/en-us/TRIZ+for+Engineers%3A+Enabling+Inventive+Problem+Solving-p-9780470741887 [Accessed 29 May 2025].

Garud, R., Tuertscher, P. & Van de Ven, A. H. 2013. Perspectives on innovation processes. Academy of Management Annals. Vol. 7, No. 1, pp. 775–819. Available at: https://doi.org/10.1080/19416520.2013.791066 [Accessed 29 May 2025].

Huston, L. & Sakkab, N. Y. 2006. Connect and develop: Inside Procter & Gamble's new model for innovation. Harvard Business Review. Vol. 84, No. 3, pp. 58–66. Available at: https://hbr.org/2006/03/connect-and-develop-inside-procter-gambles-new-model-for-innovation [Accessed 29 May 2025].

IBM. 2023. What is Robotic Process Automation (RPA)? Available at: https://www.ibm.com/think/topics/rpa [Accessed 29 May 2025].

Kolko, J. 2015. Design Thinking Comes of Age. Harvard Business Review. Vol. 93, No. 9, pp. 66–71. Available at: https://hbr.org/2015/09/design-thinking-comes-of-age [Accessed 2 October 2025].

Lacity, M. C., Willcocks, L. P., & Craig, A. 2015. Robotic process automation at Telefónica O2. Available at: https://eprints.lse.ac.uk/64516/ [Accessed 2 October 2025].

Lacity, M., & Willcocks, L. 2016. Robotic Process Automation: The Next Transformation Lever for Shared Services. Available at: https://www.umsl.edu/~lacitym/OUWP1601.pdf [Accessed 2 October 2025].

Liedtka, J. 2015. Perspective: Linking design thinking with innovation outcomes through cognitive bias reduction. Journal of Product Innovation Management. Vol. 32, No. 6, pp. 925–938. Available at: https://doi.org/10.1111/jpim.12163 [Accessed 29 May 2025].

Manthaa, A. 2019. What are the Benefits and Challenges of Implementing Robotic Process Automation (RPA) in Business? Medium. Available at: https://medium.com/@anirudh-manthaa/what-are-the-benefits-and-challenges-of-implementing-robotic-process-automation-rpa-in-business-6aa7e5cac520 [Accessed 29 May 2025].

Moehrle, M. G. 2005. What is TRIZ? From conceptual basics to a new comprehensive innovation process. Creativity and Innovation Management. Vol. 14, No. 1, pp. 3–13. Available at: https://doi.org/10.1111/j.1476-8691.2005.00320.x [Accessed 29 May 2025].

Prucha, P. 2024. Robotic Process Automation as a Driver for Sustainable Innovation and Entrepreneurship. Cornell University. Available at: https://arxiv.org/abs/2403.00431 [Accessed 29 May 2025].

Redwood. 2024. Navigating through the limitations of robotic process automation (RPA). Available at: https://www.redwood.com/article/overcoming-robotic-process-automation-limitations [Accessed 29 May 2025].

Ries, E. 2011. The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses. New York: Crown Business.

Rigby, D. K., Sutherland, J., & Takeuchi, H. 2016. Embracing Agile. Harvard Business Review. Vol. 94, No. 5, pp. 40–50. Available at: https://hbr.org/2016/05/embracing-agile [Accessed 2 October 2025].

Slade74. 2022. Stage-Gate process diagram. [Image]. Available at: https://commons.wikimedia.org/wiki/File:Stage-Gate_process.png [Accessed 2 October 2025].

Thomke, S. & Feinberg, B. 2009. Design Thinking and Innovation at Apple. Available at: https://www.hbs.edu/faculty/Pages/item.aspx?num=36789 [Accessed 2 October 2025].

UiPath. 2025. Automation Case Studies. Available at: https://www.uipath.com/resources/automation-case-studies [Accessed 29 May 2025].

van der Aalst, W.M.P., Bichler, M. & Heinzl, A. 2018. Robotic Process Automation. Bus Inf Syst Eng. Vol. 60, pp. 269-272. Available at: https://doi.org/10.1007/s12599-018-0542-4 [Accessed 2 October 2025].